

Exercise 7: Dynamic Systems
 (to be returned on Jan 8th, 8:15)

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Consider the trajectory of a some object thrown vertically into the air, in a gravity field for which we would like to estimate the constant of acceleration g . Using multiple cameras, we measure the position p of the object with measurement noise ϵ as $y = p + \epsilon$.

1. **(Continuous and Discrete Model)** From kinematics we know that the object follows the simple ordinary differential equation

$$\ddot{p} = -g.$$

To formulate a model, let p and v correspond to the vertical position and velocity of the object, respectively.

- (a) ON PAPER: By defining a state vector $x = [p, v]^\top \in \mathbb{R}^2$, we can reformulate this kinematic model into the linear affine state-space form:

$$\dot{x} = Ax + b \quad (1a)$$

$$y = Cx + d + \epsilon \quad (1b)$$

Define the values and dimensions of the matrices A and C as well of the vectors b and d accordingly. (1 points)

- (b) ON PAPER: Show that the analytical solution to the differential equation (1a), with initial state $x_0 = x(0)$,

$$x(t) = \exp(At)x_0 + \int_0^t \exp(A(t-\tau))b \, d\tau$$

is for this model equivalent to:

$$x(t) = x_0 + (Ax_0 + b)t + \frac{1}{2}Abt^2$$

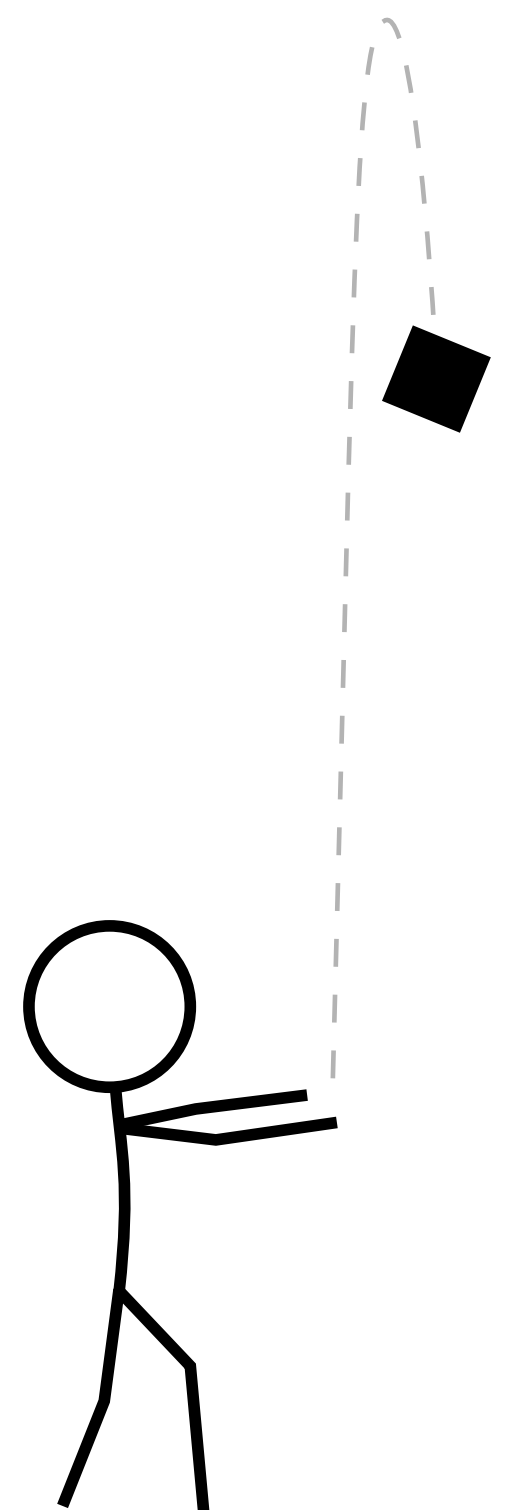
Hint: Take a close look at the high order terms of the expansion of the matrix exponential given by $\exp(At) := \mathbb{I} + At + \frac{1}{2!}(At)^2 + \frac{1}{3!}(At)^3 \dots$ (2 points)

- (c) ON PAPER: Since we later will have measurements on an equidistant time grid with stepsize h , we are interested to find a discrete state-space model

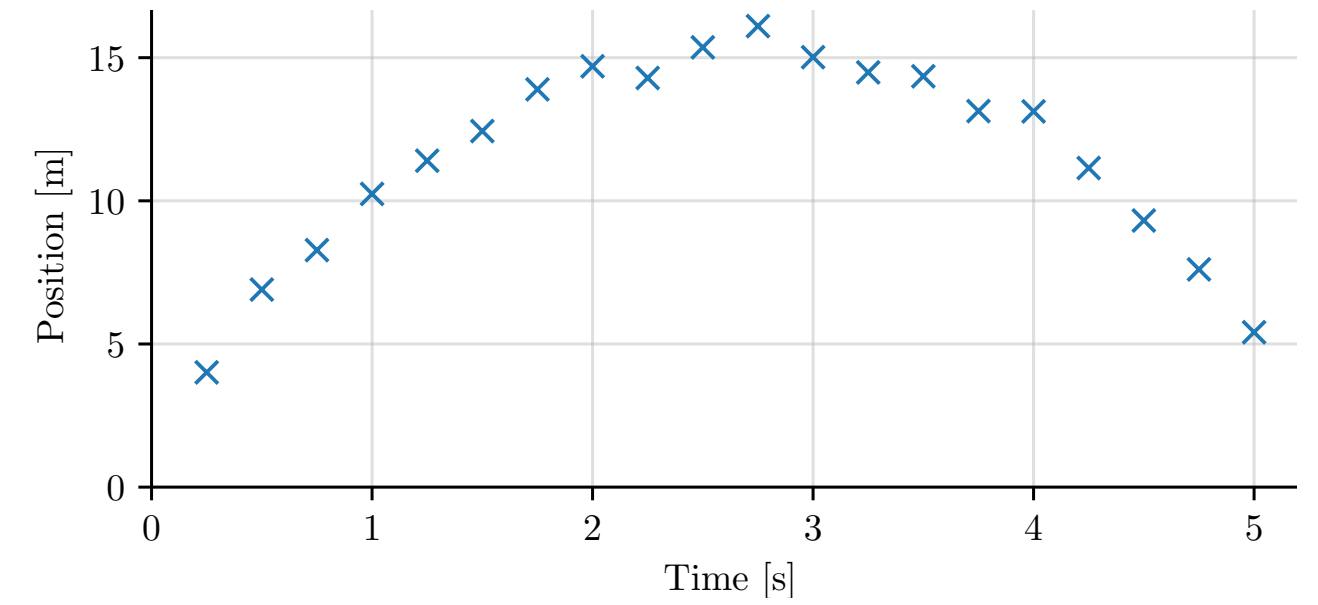
$$x_{k+1} = \tilde{A}x_k + \tilde{b} \quad (2a)$$

$$y_k = \tilde{C}x_k + \tilde{d} + \epsilon_k \quad (2b)$$

where $x_k = x(t_k)$ is the state at some sampling time $t_k = kh$ and $x_{k+1} = x(t_{k+1}) = x(t_k + h)$ is the state at the next sampling time. Use the equations of the analytical solution above to define the matrices and vectors $\tilde{A}, \tilde{b}, \tilde{C}$ and $\tilde{d}$ of the discrete model. (2 points)



2. We assume that the position measurements (the output of our model) are subject to i.i.d. noise ϵ_k with a standard deviation of $\sigma_\epsilon = 0.3$ m. The series of $N = 20$ measurements $y_1, y_2, \dots, y_N$ lies on an evenly spaced time grid $t_k = kh, k = 1, \dots, N$, with $h = 0.25$ s.



(Physical Model) From the solution of the previous tasks we can derive a model for the measurements that is inspired from physics as

$$y_k = p_0 + v_0 t_k - \frac{1}{2} g t_k^2 + \epsilon_k \quad (3)$$

that is parameterized by the initial position $p_0 = p(t_0)$, initial velocity $v_0 = v(t_0)$ and the constant g .

- (a) ON PAPER: To estimate the unknown parameters $\theta = [p_0, v_0, g]^\top \in \mathbb{R}^3$ we want use linear least squares with the measurement model above and thus want to solve the problem

$$\theta_{\text{phy}} = \arg \min_{\theta} \|y_{\text{phy}} - \Phi_{\text{phy}} \theta\|_2^2.$$

Define the vectors y_{phy} and the matrix Φ_{phy} , along with their dimensions. (0.5 points)

- (b) CODE: Use linear least squares to find the solution to the estimation problem. Estimate the covariance matrix Σ_{θ} of the estimation result. Complete the code to plot the trajectory of the predicted output with the found parameters over time, along with a 2-sigma confidence interval of the predicted output y . For this you will need to calculate the covariance of the predicted outputs using the formula

$$\Sigma_{y_{\text{phy}}} = \Phi_{\text{phy}} \Sigma_{\theta} \Phi_{\text{phy}}^\top + \Sigma_{\epsilon} \quad (2 \text{ points})$$

- (c) PAPER: Where is the experiment taking place? (0 points)

(ARX Model) Another possible modelling choice is to use an autoregressive model

$$y_{k+1} = \theta_1 y_{k-1} + \theta_2 y_k + \theta_3 \quad (4)$$

that relates the next output to previous outputs and a constant. With this model, we can only estimate the constant g , not the initial position and velocity.

- (d) ON PAPER: Show how the discrete linear state-space model (2a) can be reformulated into the ARX model (4). How does the constant g relate to the coefficients $\theta = [\theta_1, \theta_2, \theta_3]^\top$? (1 points)
- (e) ON PAPER: Fit an ARX model to the data by formulating an LLS problem

$$\theta_{\text{ARX}} = \arg \min_{\theta} \|y_{\text{ARX}} - \Phi_{\text{ARX}} \theta\|_2^2.$$

Define the vectors y_{ARX} and the matrix Φ_{ARX} , along with their dimensions. (0.5 points)

- (f) CODE: Use LLS to find the coefficients of the ARX model. Use the code in the template to visualize a ‘rollout’ of the model. (0.5 points)
- (g) ON PAPER: Why do you think the ARX modelling approach performs so much worse? (0.5 points)

This sheet gives in total 10 points

1). p: vertical pos. $\downarrow$ $x = [p, v]^T \in \mathbb{R}^{2 \times 1}$
 v : vel. $\downarrow$ $\dot{x} = [\dot{p}, \dot{v}]^T \rightarrow \dot{v} = g$

Given $\ddot{p} = -g$

$\ddot{p}$ is the acceleration.

a) Turn into state-space model.

$$\begin{bmatrix} \dot{p} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ -g \end{bmatrix}$$

$$\dot{x} = Ax + b$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p \\ v \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix}$$

$$y = Cx + d + e$$

b). Given $\exp(At) = I + At + \frac{1}{2!}(At)^2 + \frac{1}{3!}(At)^3 + \dots \underbrace{O(n)}$

with matrix $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow A^n = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \dots \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$n > 1$$

$\Rightarrow$ the higher order term will disappear, leading to

$$\exp(At) = I + At.$$

• The first term: $\exp(At) \cdot x_0 = (I + At) \cdot x_0$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \cdot t \right) \cdot \begin{bmatrix} p_0 \\ v_0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} p_0 + v_0 t \\ v_0 \end{bmatrix}$$

• The term: $\exp(A(t-\tau)) = I + A(t-\tau) + \frac{1}{2!}(A(t-\tau))^2 + \dots$
 $\quad \quad \quad = I + A(t-\tau)$

$$\Rightarrow \exp(A(t-\tau)) \cdot b = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} (t-\tau) \right) \begin{bmatrix} 0 \\ -g \end{bmatrix}$$

$$= \begin{bmatrix} 1 & t-\tau \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -g \end{bmatrix} = \begin{bmatrix} (t-\tau)g \\ -g \end{bmatrix}$$

$$= -g \begin{bmatrix} (t-\tau) \\ 1 \end{bmatrix}$$

$$\rightarrow \int_0^t \exp(A(t-\tau)) \cdot b \cdot d\tau = \int_0^t \begin{bmatrix} -g(t-\tau) \\ -g \end{bmatrix} d\tau$$

$\nearrow -g\tau \Big|_0^t + g \frac{\tau^2}{2} \Big|_0^t$
 $\searrow -g\tau \Big|_0^t$

$$= \begin{bmatrix} -gt^2 + \frac{1}{2}gt^2 \\ -gt \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}gt^2 \\ -gt \end{bmatrix}$$

$$\Rightarrow x(t) = \begin{bmatrix} p_0 + v_0 t \\ v_0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2}gt^2 \\ -gt \end{bmatrix}$$

$$= \begin{bmatrix} p_0 + v_0 t - \frac{1}{2}gt^2 \\ v_0 - gt \end{bmatrix} \quad (a)$$

• the $x(t) = r_0 + (A r_0 + b)t + \frac{1}{2} A b t^2$

$$= \begin{bmatrix} p_0 \\ v_0 \end{bmatrix} + \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_0 \\ v_0 \end{bmatrix} + \begin{bmatrix} 0 \\ -g \end{bmatrix} \right) t + \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -g \end{bmatrix} t^2$$

$$= \begin{bmatrix} p_0 \\ v_0 \end{bmatrix} + \begin{bmatrix} v_0 t \\ -gt \end{bmatrix} + \begin{bmatrix} -\frac{1}{2}gt^2 \\ 0 \end{bmatrix}$$

$\underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_0 \\ v_0 \end{bmatrix}}_{\begin{bmatrix} v_0 \\ 0 \end{bmatrix}}$
 $\underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -g \end{bmatrix}}_{\begin{bmatrix} -g \\ 0 \end{bmatrix}}$

$$= \begin{bmatrix} p_0 + v_0 t - \frac{1}{2}gt^2 \\ v_0 - gt \end{bmatrix} \quad (b)$$

(a) = (b) $\rightarrow$ proved.

$$c). \quad x_{k+1} = \tilde{A} x_k + \tilde{b}$$

$$y_k = \tilde{C} x_k + \tilde{d} + \epsilon_k$$

$$x(t) = x_0 + (Ax_0 + b)t + \frac{1}{2} Abt^2$$

$$\rightarrow x_{k+1} = x_k + (Ax_k + b)h + \frac{1}{2} Abh^2$$

$$= (I + Ah)x_k + \underbrace{bh + \frac{1}{2} Abh^2}$$

$$= (I + Ah)x_k +$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} h \right) x_k + \begin{bmatrix} 0 \\ -g \end{bmatrix} \cdot h + \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -g \end{bmatrix} \cdot h^2$$

$$= \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ -gh \end{bmatrix} + \begin{bmatrix} -1/2 gh^2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} p_{k+1} \\ v_{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix}}_{\tilde{A}} \underbrace{\begin{bmatrix} p_k \\ v_k \end{bmatrix}}_{\tilde{b}} + \underbrace{\begin{bmatrix} -1/2 gh^2 \\ -gh \end{bmatrix}}_{\tilde{b}}$$

$$y_k = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\tilde{C}} \underbrace{\begin{bmatrix} p_k \\ v_k \end{bmatrix}}_{\tilde{d}} + \epsilon_k$$

• This method is called exact discretization $\rightarrow$ it is good for LTI system. For the complex non-linear or linear time variant system, this method is not suitable $\rightarrow$

1. Euler approximation. $\dot{x} \approx \frac{x_{k+1} - x_k}{h}$

$$\rightarrow \frac{x_{k+1} - x_k}{h} = Ax_k + b$$

$$\rightarrow x_{k+1} = (I + hA)x_k + hb.$$

2. RK4.

Q2 $y_k = p_0 + v_0 t_k - \frac{1}{2} g t_k^2 + \varepsilon_k$

a) $y_k = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N$

$$\Phi_{phy} = \begin{bmatrix} 1 & t_1 & -\frac{1}{2}t_1^2 \\ 1 & t_2 & -\frac{1}{2}t_2^2 \\ \vdots & \vdots & \vdots \\ 1 & t_N & -\frac{1}{2}t_N^2 \end{bmatrix} = \begin{bmatrix} 1 & h & -\frac{1}{2}h^2 \\ 1 & 2h & -\frac{1}{2}(2h)^2 \\ \vdots & \vdots & \vdots \\ 1 & Nh & -\frac{1}{2}(Nh)^2 \end{bmatrix} \in \mathbb{R}^{N \times 3}$$

$$\theta_{phy} = \underset{\theta}{\operatorname{argmin}} \|y_{phy} - \Phi_{phy} \theta\|_2^2$$

$$\rightarrow \theta^*_{phy} = (\Phi_{phy}^T \Phi_{phy})^{-1} \Phi_{phy}^T y_{phy}$$

$$y_{phy} = \Phi_{phy} \theta + \varepsilon$$

$$\text{from } \theta^*_{phy} = (\Phi_{phy}^T \Phi_{phy})^{-1} \Phi_{phy}^T y_{phy}$$

$$\rightarrow \operatorname{cov}(\theta^*_{phy}) = \operatorname{cov}(A \cdot y_{phy})$$

$$= A \cdot \operatorname{cov}(y_{phy}) A^T$$

Given the $\sigma_\varepsilon = 0.3$ is i.i.d (identical and independent distributed).

$$\Rightarrow \text{cov}(y_{\text{phy}}) = \sigma_e^2 \cdot \mathbb{I} \quad (\mathbb{I} \in \mathbb{R}^{3 \times 3})$$

$$\Rightarrow \text{cov}(\theta_{\text{phy}}^*) = A \cdot \sigma_e^2 \cdot \mathbb{I} \cdot A^T$$

$$A = (\Phi_{\text{phy}}^T \Phi_{\text{phy}})^{-1} \Phi_{\text{phy}}^T$$

$$= \sigma_e^2 (\Phi_{\text{phy}}^T \Phi_{\text{phy}})^{-1} \Phi_{\text{phy}}^T [(\Phi_{\text{phy}}^T \Phi_{\text{phy}})^{-1} \Phi_{\text{phy}}^T]^T$$

$$= \sigma_e^2 (\Phi^T \cdot \Phi)^{-1} \Phi^T \cdot \Phi \underbrace{((\Phi^T \cdot \Phi)^{-1})^T}_{\text{symmetric matrix}}$$

$$=$$

$$= \sigma_e^2 (\Phi^T \cdot \Phi)^{-1} \underbrace{\Phi^T \cdot \Phi}_{\text{square}} \underbrace{(\Phi^T \cdot \Phi)^{-1}}_{\text{Inverse of square}}$$

$\mathbb{I}$

$$= \sigma_e^2 (\Phi^T \cdot \Phi)^{-1} \cdot \mathbb{I} = \boxed{\sigma_e^2 (\Phi^T \cdot \Phi)^{-1}}$$

c) $\Rightarrow g = 3.708 \text{ m/s}^2 \Rightarrow$ The experiment was taken on mercury

d) $y_{k+1} = \theta_1 y_{k-1} + \theta_2 y_k + \theta_3$ (ARX model)

given: $p_{k+1} = p_k + v_k h - \frac{1}{2} g h^2$

since $y_k = p_k \Rightarrow y_{k+1} = y_k + v_k h - \frac{1}{2} g h^2$

$v_k = v_{k-1} - g h$

$$\Rightarrow y_{k+1} = y_k + (v_{k-1} - g h) \cdot h - \frac{1}{2} g h^2$$

$$y_k = y_{k-1} + v_{k-1} \cdot h - \frac{1}{2} g h^2 \rightarrow \underline{v_{k-1} \cdot h = y_k - y_{k-1} + \frac{1}{2} g h^2}$$

$$\bullet y_{k+1} = y_k + \underline{v_{k-1}} \cdot h - gh^2 - \frac{1}{2} gh^2$$

$$= y_k + y_k - y_{k-1} + \cancel{\frac{1}{2} gh^2} - gh^2 - \cancel{\frac{1}{2} gh^2}$$

$$y_{k+1} = 2y_k - y_{k-1} - gh^2$$

$$= -y_{k-1} + 2y_k - gh^2$$

$$\Rightarrow \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -gh^2 \end{bmatrix} \Rightarrow g = \frac{-\theta_3}{h^2}$$

ARX model: $y_{k+1} = -y_{k-1} + 2y_k - gh^2, k=1, \dots, N$

$$y_{ARX} = \begin{bmatrix} y_3 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^{N-2}; \quad \Phi_{ARX} = \begin{bmatrix} y_1 & y_2 & +1 \\ y_2 & y_3 & +1 \\ \vdots & \vdots & \vdots \\ y_{N-2} & y_{N-1} & +1 \end{bmatrix} \in \mathbb{R}^{(N-2) \times 3}$$

• The result of this approach is worse than using the physical model.

→ Because, without the dictation of physical model the estimator is tricked by the noise data (σ_ϵ), yielding the error in θ^* . This leads to the degradations in the roll out ARX.

Opened. Known $\theta = \begin{bmatrix} -1 \\ 2 \\ \theta_3 \end{bmatrix} \Rightarrow \theta = [\theta_3]$

$$y_{k+1} = -y_{k-1} + 2y_k - gh^2$$

$$\Rightarrow y_{k+1} + y_{k-1} - 2y_k = \underline{\underline{-gh^2}}$$

formulate the problem again with known $\theta_1, \theta_2 \rightarrow$ solve for θ_3

$$y_{ARX} = \begin{bmatrix} y_3 & y_1 & -2y_2 \\ y_4 & y_2 & -2y_3 \\ \vdots & \vdots & \vdots \\ y_N & y_{N-2} & -2y_{N-1} \end{bmatrix} \quad \bigg| \quad \Phi_{ARX} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^{(N-2)}$$

$$\in \mathbb{R}^{(N-2) \times 3}$$

If try to formulate $y_{ARX} \in \mathbb{R}^{(N-2) \times 3}$

$\rightarrow$ The result will fault, because y_{ARC} must be a vector

$$y_{k+1} + y_{k-1} - 2y_k$$

$\rightarrow$ This term must be mapped into

$$y_{ARX} = [y[2:N] + y[0:N-2] - 2 \cdot y[1:N-1]]$$

$\rightarrow$ The result becomes better than, but the error from the measurements makes the solution of the prediction become worse and worse when the num. of step increase.

$\rightarrow$ Adding a filter, where

$$y_{pred} = y_{pred} + K \cdot (y_{meas} - y_{pred})$$

$\rightarrow K \in [0; 1]$ the increase of K means more trust on measurements and vice versa.